



1. a) Consider the system with input x and output y described by the differential equation:

$$\frac{d^2y}{dt^2} = ay + bx$$

Give the Laplace transform of the response to initial conditions, or zero-input response.

- b) Give the time-domain function $y_{zi}(t)$ for $a = -1$ and for $a = 1$.
2. a) By means of the Laplace transform, find the solutions $x_1(t)$ and $x_2(t)$ of the following system:

$$\begin{aligned}\frac{dx_1(t)}{dt} &= x_2 \\ \frac{dx_2(t)}{dt} &= -2x_1 - 3x_2 + u(t),\end{aligned}$$

where $x_1(0) = 1$, $x_2(0) = 0$, and $u(t)$ is a step input of magnitude 1.

- b) Write a state-space realization for the following system and give the values of its poles.

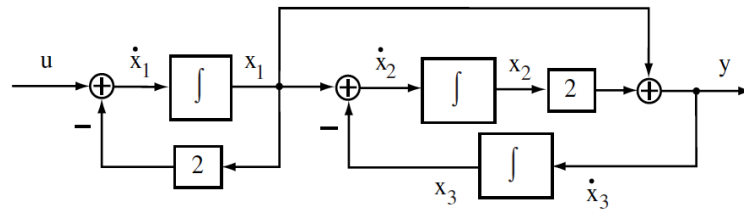


Figure 1: Question-1

3. a) Determine what the steady state output $y_{ss}(t)$ is for an input $x(t) = \sin(t)$ and a system with transfer function.

$$H(s) = \frac{-(s-1)}{(s+1)^2}$$

- b) For the system and input signal of part (a), use partial fraction expansions to determine the transient response $y_{tr}(t)$ (only the transient response is needed).
4. a) Indicate whether the following system is BIBO stable.

$$H(s) = \frac{s-1}{(s+2+j)^2(s+2-j)^2}$$

- b) Indicate whether the following system is BIBO stable.

$$H(s) = \frac{s+1}{(s+j)(s-j)}$$

- c) Provide a signal for which the following transform is (i) bounded, and (ii) converges.

$$H(s) = \frac{s^2 + 9}{(s^2 - 1)(s^2 + 4)}$$

- d) A signal $x(t) = \cos(4t)$ is applied to a system with transfer function $H(s) = 1/(s^2 + 4)$. Is the output bounded?
- e) A signal that converges to zero is applied to a BIBO stable system. The transform of the signal and the transfer function of the system are proper, rational functions of s . Indicate whether the output of the system: (i) is bounded, (ii) converges, and (iii) converges to zero.
5. a) Find the transfer function $H(s) = Y(s)/X(s)$ for the following system

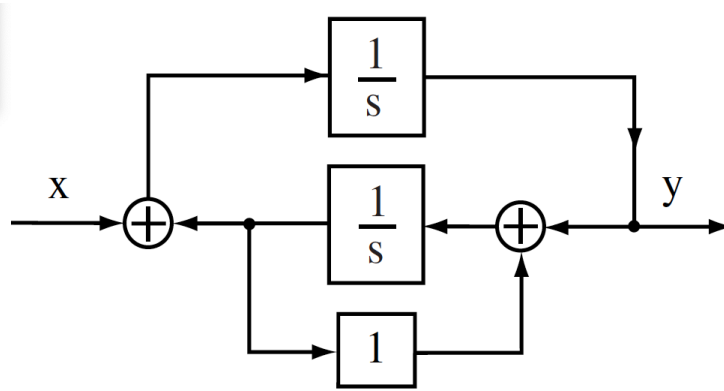


Figure 2: Question-5

- b) What can you say about the steady-state value of the output $y_{ss}(t)$ for a constant input $x(t) = 5$.
6. Show that every system (A, b) in the following form is controllable for any values of a_i 's, $i = 0, 1, \dots, n-1$.

$$A = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 0 & 0 & \ddots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ -a_0 & -a_1 & \dots & -a_{n-1} \end{bmatrix}, b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

7. i) State a choice for the matrix B such that the pair (A, B) becomes controllable where:

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

ii) For what value of a will the following pair be controllable:

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$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}, B = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$$

8. Show that if the system (A, B) is controllable, then $(\bar{A}, \bar{B})$ is also controllable where $\bar{A} = T^{-1}AT$ and $\bar{B} = T^{-1}B$ for some non-singular matrix T . Prove using the two tests for controllability. Show vice versa as well. (Hint: use the fact that $\text{rank}(PQ) = \text{rank}(Q)$ when P is non-singular and $\text{rank}(PQ) = \text{rank}(P)$ when Q is non-singular).